

Landing Kinematics and Isometric Hip Strength of Individuals With Chronic Ankle Instability

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Abstract

Background: Chronic ankle instability (CAI) is associated with hip strength deficits and altered movement in the lower extremity. However, it remains unclear how hip strength deficits contribute to lateral ankle sprain (LAS) mechanisms. We aimed to compare lower extremity landing kinematics and isometric hip strength between individuals with and without CAI and examine associations between hip kinematics and strength.

Methods: Seventy-six individuals completed 5 single-leg landings, during which we collected three-dimensional ankle, knee, and hip kinematics from 200 milliseconds pre-initial contact to 50 milliseconds post-initial contact. We calculated average peak torque (Nm/kg) from 3 trials of isometric hip extension, abduction, and external rotation strength testing. One-way analyses of variance assessed group differences (CAI, LAS copers, and control) in hip strength and kinematics. Pearson product moment correlations assessed associations between hip kinematics and strength. We adjusted the kinematic group comparisons and correlation analyses for multiple comparisons using the Benjamini-Hochberg method.

Results: The CAI group exhibited less hip abduction during landing than LAS copers and controls. The CAI group had lower hip external rotation strength than LAS copers ($P = .04$, $d = 0.62$ [0.05, 1.17]) and controls ($P < .01$, $d = 0.87$ [0.28, 1.43]). Effect sizes suggest that the CAI group had deficits in EXT compared with controls ($d = 0.63$ [0.06, 1.19]). Hip strength was not associated with hip landing kinematics for any group.

Conclusion: Altered landing mechanics displayed by the CAI group may promote mechanisms of LAS, but they are not associated with isometric hip strength. However, hip strength deficits may negatively impact other functional tasks, and they should still be considered during rehabilitation.

Level of Evidence: Level III, case-control study.

Keywords: ankle, hip musculature, motion capture, handheld dynamometry

Introduction

Ankle sprains are an extremely common musculoskeletal pathology, accounting for an estimated 600 000 to 1 million emergency department visits in the United States annually.^{37,44} Lateral ankle sprains (LASs), in particular, are of significant concern due to vast rates of recurrence, which surpass those of any other lower extremity musculoskeletal injury.¹⁸ Chronic ankle instability (CAI) commonly develops after a LAS and is characterized by recurrent LAS, feelings of instability, and “giving-way” episodes.^{19–21} The rate of CAI development after a LAS is estimated to be as high as 74%.^{1,5} Conversely, LAS copers are able to avoid mechanisms of recurrent injury after an initial LAS.⁴⁶

The hallmark mechanism of an index and recurrent LAS is excessive rearfoot supination,¹⁶ composed of ankle plantarflexion, subtalar inversion, and internal

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rotation of the foot,⁴⁵ as transition from nonweightbearing to weightbearing takes place. It has been speculated that these combined movement patterns increase loading under the lateral column of the foot,^{16,24,25} potentially creating external supination moments at the ankle and stressing the lateral ankle ligaments.¹⁶ While foot placement is partially controlled by ankle joint motion, neuromuscular control of proximal joints likely has meaningful contributions as well. In particular, weaker posterolateral hip musculature and medial, internally rotated lower extremity positioning appear to promote LAS mechanisms.^{4,12,28,29} Individuals with CAI have previously exhibited such characteristics, potentially indicating that centrally mediated neuromuscular impairments are contributing to recurrent injury.⁴³

A number of investigators have documented altered three-dimensional (3D) hip, knee, and ankle kinematics during landing tasks in individuals with CAI compared with LAS copers and individuals without a previous LAS.^{6-8,10,13,14,17,22} Some of the differences identified between CAI and healthy control groups include greater knee flexion, greater ankle dorsiflexion, greater ankle inversion, and less hip external rotation (ER), although these findings were inconsistent. A few kinematic analyses have also produced mixed results regarding 3D ankle and knee as well as frontal and transverse plane hip kinematics when comparing CAI and LAS coper groups.^{6,7,13,14} Isometric hip strength deficits of the posterolateral hip musculature have further exemplified proximal neuromuscular changes in individuals with CAI compared with LAS copers and controls.^{15,31,33} While reduced hip abduction (ABD) strength is associated with reduced dynamic postural control of individuals with CAI, the influence of posterolateral hip muscular strength over landing patterns has not been explored in these populations. Consequently, it remains unclear how proximal neuromuscular changes contribute to mechanisms of recurrent LAS and giving-way episodes that plague those with CAI. If altered lower extremity landing patterns are influenced by hip muscular strength deficits in those with CAI, expanded rehabilitation protocols that target proximal joint motor control may be needed to mitigate LAS mechanisms.

Therefore, the purpose of this study was to compare differences in hip muscular strength and 3D lower extremity kinematics during a single-leg landing task between individuals with CAI, LAS copers, and healthy controls. We selected a single-leg landing task followed by stabilization as this has been a commonly used method of assessing landing kinematics in CAI cohorts.^{10,13,17,22} Additionally, we aimed to determine the association of hip muscular strength with hip kinematics. We hypothesized that individuals with CAI would display lower hip muscular strength, and altered ankle, knee, and hip kinematics compared with individuals without CAI. Furthermore, we

hypothesized that in all cohorts, decreased posterolateral hip muscular strength would be associated with lower degrees of hip flexion, ABD, and ER during landing.

Methods

Participants

Seventy-six participants between the ages of 18 and 35 (female: 56, male: 20; mean [SD]: 23.7 [4.3] years; 167.6 [7.9] cm; 70.7 [15.2] kg) from the surrounding university community volunteered for the study. This study was part of a larger exploration of impairments of individuals with CAI, and we determined the sample size needed from hip strength and balance data of previous studies.^{15,35} We estimated that 75 total participants (25 per group) were needed to find statistically significant differences at a predetermined alpha level of 0.05 and power of 0.80. Individuals with lower extremity injuries aside from a LAS in the previous 2 years and any history of lower extremity fracture or surgery were excluded from participation. A single member of the research team assigned participants to CAI, LAS coper, and control groups utilizing previously established criteria.^{19-21,46}

Inclusion criteria for the CAI group consisted of 1 or more previous LASs resulting in inflammatory symptoms and 1 or more days of activity time loss. At least 12 months must have passed between the initial LAS and study enrollment, and no recurrent LAS could have occurred within 3 months prior to study enrollment. Members of the CAI group also must have reported 2 or more repeated episodes of giving way, feelings of instability, or recurrent LAS within 5 months of study enrollment. Additionally, participants with CAI answered yes to 5 or more questions on the Ankle Instability Instrument (AII), scored 11 or greater on the Identification of Functional Ankle Instability (IdFAI), and scored 24 or lower on the Cumberland Ankle Instability Tool (CAIT). If a participant had a previous LAS bilaterally, we tested the limb with a greater number of giving-way episodes and lower scores on the Foot and Ankle Ability Measure Activities of Daily Living (FAAM-ADL) and Sport (FAAM-S) subscales.³¹

Inclusion criteria for the LAS coper group consisted of 1 previous LAS resulting in inflammatory symptoms and 1 or more days of activity time loss. At least 12 months must have passed between the initial LAS and study enrollment. Members of the LAS coper group also must have reported no episodes of giving way, feelings of instability, or recurrent LAS. Additionally, participants in the LAS coper group answered yes to 4 or fewer questions on the AII, scored 10 or less on the IdFAI, and scored 24 or greater on the CAIT. Lastly, participants in the LAS coper group self-reported that they had not modified physical and occupational activity levels due to their ankle injury.

Inclusion criteria for the control group consisted of no previous LAS and no previous episodes of giving way or feelings of instability in their ankles. Additionally, participants in the control group answered no to all questions on the AII, scored 0 on the IdFAI, and scored 30 on the CAIT.

Instrumentation

A Vertec vertical jump tester (Sports Imports, Columbus, OH) measured maximum vertical jump height and provided a target during the jump-landing task. Five Eagle digital cameras, 1 Eagle-4 digital camera, and 4 Raptor-E digital cameras (Motion Analysis Corporation, Santa Rosa, CA) were used to collect kinematic data. Fifty-six retro-reflective markers fixed to specific anatomical landmarks (bilateral markers on iliac crests, anterior superior iliac spines, posterior iliac spines, greater trochanters, lateral thighs [$\times 4$], anterior distal thighs, medial and lateral femoral epicondyles, patellae, tibial tuberosities, lateral shanks [$\times 4$], anterior ankles, medial and lateral malleoli, posterior calcanei, navicular tuberosities, base of fifth metatarsals, head of first and second metatarsals, and acromioclavicular joints, and single markers on the sacrum, seventh cervical spinous process, manubrium, and inferior angle of the right scapula) were tracked by the 10 digital cameras. A Bertec force plate (Bertec FP6090-15-2000, Bertec Inc., Columbus, OH) embedded in the laboratory floor was used to collect kinetic data required for event determination. Cortex 5.5 motion capture and processing software (Motion Analysis Corporation, Santa Rosa, CA) was used to capture all kinematic and kinetic data. Kinematic data were sampled at a rate of 200 Hz, and kinetic data were sampled at a rate of 1000 Hz. A handheld dynamometer assembled from a portable load cell and designated attachments (BTE Evaluator, BTE, Hanover, MD) were used to measure isometric hip strength.

Procedures

All study procedures were conducted in a single visit to the research laboratory. Each participant first read and signed a written informed consent document approved by the university institutional review board. Prior to measuring primary outcomes, participant demographics were collected, and each participant was permitted 5 minutes for self-selected warm-up and stretching. Participants in CAI and LAS coper groups underwent testing on their involved limb, and members of the control group underwent testing on a randomly selected limb. Members of the research team conducting laboratory tests were blinded to participants' group membership.

Prior to collecting motion capture data, the maximum vertical jump height was determined from the best of 3 double-leg vertical jumps, using the Vertec vertical jump

tester. Next, the retro-reflective markers were applied, and participants completed a static stance trial to create a kinematic model of the skeletal segments. For the single-leg landing task, participants assumed a double-leg stance 70 cm from the center of the force plate. Participants jumped from 2 feet in a forward direction, reaching with 1 hand to touch the Vertec vane set to 50% of their maximum vertical jump height.³⁶ Participants were required to land on the force plate with the designated test limb and maintain a single-leg stance with arms folded across their chests for 5 seconds.³⁰ Participants were allotted 4 practice trials, and then repeated the task until 5 successful trials were obtained. Criteria for failed trials consisted of missing the Vertec target, missing the force plate entirely or partially upon landing, losing contact with the force plate between initial contact (IC) and 5 seconds postlanding, and touching down with the nontest limb.

We measured lower extremity segment lengths to represent moment arms for hip strength normalization. The femur length was measured from the center of the greater trochanter to the lateral femoral epicondyle. The tibia length was measured from the medial knee joint line to the distal end of the medial malleolus. In random order, we measured isometric hip extension (EXT), ABD, and ER strength following procedures described by Thorborg et al.⁴² Participants were prone on a plinth with the knee of the test limb flexed to 90 degrees during EXT assessment. Stabilization straps were placed over the participants' posterior superior iliac spines and the dynamometer, which was positioned 5.08 cm proximal to the knee joint line. Participants lied on the side of the nontest limb during ABD assessment, with approximately 10 degrees of hip ABD maintained with pillows placed between the participants' knees. Stabilization straps were placed over the participants' iliac crest and the dynamometer, which was positioned 5.08 cm proximal to the knee joint line. Participants sat with their hips and knees flexed to 90 degrees, lower legs hanging off the edge of the table, and arms folded across their chest during ER assessment. A stabilization strap was placed across the participants' thighs, and the dynamometer was positioned 5.08 cm proximal to the distal end of the medial malleolus. During strength assessments, participants pushed against the dynamometer for 5 seconds. Participants were instructed to gradually increase force for the first 3 seconds, and then exert maximum force during the last 2 seconds. A single practice trial preceded 3 test trials. Rest intervals between trials lasted 30 seconds. We recorded peak external force (N) for each trial as the variable of interest. We also examined the test trials' coefficient of variation, and if it exceeded 10.0, the trial with the greatest deviation from the mean was repeated. Test-retest reliability is reportedly good to excellent for EXT (intraclass correlation coefficient [ICC] = 0.98), ABD (ICC = 0.76), and ER (ICC = 0.95).⁴²

Table 1. Between-Group Comparisons of Primary Demographics.

	CAI, Mean $\pm$ SD ($n = 26$)	LAS Coper, Mean $\pm$ SD ($n = 25$)	Control, Mean $\pm$ SD ($n = 25$)	One-Way ANOVA F Value	P Value
Age, y	24.2 $\pm$ 4.0	24.0 $\pm$ 5.2	23.0 $\pm$ 3.6	$F_{(2, 73)} = 0.561$	.573
Height, cm	167.2 $\pm$ 7.3	166.7 $\pm$ 8.8	168.7 $\pm$ 7.6	$F_{(2, 73)} = 0.431$	.652
Mass, kg	73.5 $\pm$ 14.9	69.5 $\pm$ 17.1	69.1 $\pm$ 13.6	$F_{(2, 73)} = 0.630$	.536

Abbreviations: ANOVA, analysis of variance; CAI, chronic ankle instability; LAS, lateral ankle sprain; SD, standard deviation.

Data Processing

Raw motion capture trials collected during the single-leg landing task were imported to Visual 3D (version 5, C-Motion, Germantown, MD) for the processing of kinematic and kinetic data. Kinematic data were filtered at 6 Hz and raw GRF data were filtered at 12 Hz with a fourth-order low-pass Butterworth filter. For each raw trial, a kinematic model including trunk, pelvis, bilateral thigh, shank, and foot segments was created. Trunk and pelvis segments were oriented relative to the laboratory coordinate system, and lower extremity segments were defined locally. The hip joint center location was predicted using combined methodologies described previously.² The knee joint center location was defined by the radius between medial and lateral femoral epicondyle markers.^{38,41} The ankle joint center location was defined by the radius between medial and lateral malleoli markers.⁹ An x-y-z Cardan angle sequence was utilized to calculate hip, knee, and ankle rotations relative to each participant's neutral stance trial. Kinematic data were upsampled to match the sampling rate of kinetic data (1000 Hz). Time-averaged kinematic data were calculated for each participant from 200 milliseconds pre-IC to 50 milliseconds post-IC. We defined IC as the instance in which the force plate exceeded 10 N of force.

The peak torque of each isometric hip strength measure was averaged across 3 trials, multiplied by the length of the corresponding moment arm (m) and divided by body mass (kg) to obtain normalized torque (Nm/kg). The femur length represented the moment arm for EXT and ABD, and the tibia length represented the moment arm for ER.

Statistical Analysis

All statistical analyses were conducted using IBM SPSS Statistics version 23 (IBM Corporation, Armonk, NY). Significance levels were set a priori at $P < .05$. We tested for normality of hip strength and kinematic data using Shapiro-Wilk tests. With the exception of the CAI group's ER strength data ($P < .01$), all data were normally distributed ($P > .5$). We also tested for equal variances of hip strength and kinematic data using Levene's tests. With the exception of frontal plane hip kinematics from IC to 19 milliseconds post-IC and sagittal plane knee kinematics from 137 milliseconds pre-IC to 38 milliseconds post-IC, all data

had equal variances between groups ($P > .05$). Therefore, separate 1-way ANOVAs were used to assess group differences in primary demographics and sagittal, frontal, and transverse plane kinematics of the hip, knee, and ankle joints at each time point (milliseconds) from 200 milliseconds pre-IC to 50 milliseconds post-IC. We adjusted the kinematic analyses for multiple comparisons using the Benjamini-Hochberg method with a false discovery rate of 10%.³ Pairwise comparisons were made with Fisher's least significant difference (LSD) in the event of a significant main effect. Isometric hip muscular strength was compared between groups using separate 1-way ANOVAs, Fisher's LSD, and Cohen's d effect sizes (small: $0.20 \leq d < 0.50$, moderate: $0.50 \leq d < 0.80$, large: $d \geq 0.80$) with 95% confidence intervals (CIs). Separate Pearson product moment correlations examined associations between EXT strength and sagittal plane hip kinematics, ABD strength and frontal plane hip kinematics, and ER strength and transverse plane hip kinematics at each millisecond for each group separately. We also adjusted the correlation analyses for multiple comparisons using the Benjamini-Hochberg method with a false discovery rate of 10%.³

Results

No significant between-group differences were present for age, height, and mass (Table 1). Time-averaged 3D lower extremity kinematics with standard error are presented in Figures 1 to 3. Significant group differences existed for frontal plane hip kinematics from 78 milliseconds pre-IC to 34 milliseconds post-IC (Figure 1). The CAI group displayed significantly less hip ABD than the LAS coper and control groups across that interval. No differences in frontal plane hip angle existed between the LAS coper and control groups. No significant main effects were present for sagittal and transverse plane hip kinematics (Figure 1), 3D knee kinematics (Figure 2), or 3D ankle kinematics (Figure 3).

Between-group comparisons of isometric hip muscular strength measures are presented in Table 2. The CAI group exhibited significantly lower ER compared with the LAS coper ($P = .040$) and control ($P = .002$) groups. The effect sizes indicate that the CAI group had a moderate to large deficit in ER compared with the LAS coper ($d = 0.62$

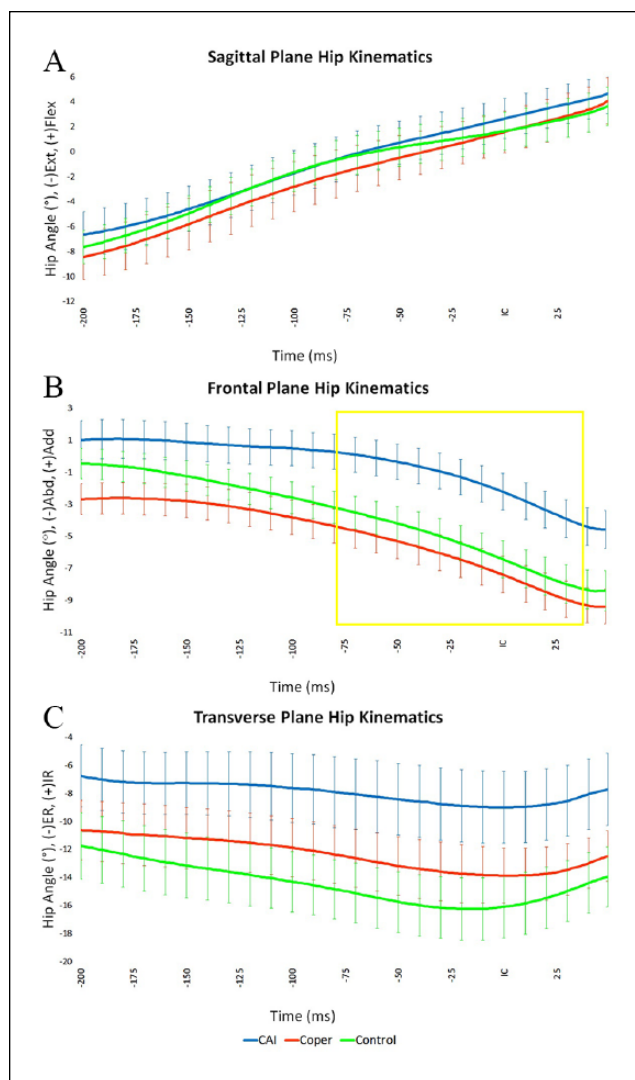


Figure 1. Pre- and postlanding hip kinematics of CAI, chronic ankle instability (CAI), lateral ankle sprain (LAS) coper, and control groups. Group means for hip kinematics in (A) sagittal, (B) frontal, and (C) transverse planes from 200 milliseconds pre-initial contact to 50 milliseconds post-initial contact. Error bars represent the standard error of the mean. The box represents the time intervals with the significant main effects for the groups ($P < .05$). Abd, abduction; Add, adduction; ER, external rotation; Ext, extension; Flex, flexion; IR, internal rotation.

[0.05, 1.17]) and control groups ($d = 0.87$ [0.28, 1.43]). No statistically significant differences existed for EXT or ABD, but the effect size suggests that the CAI group had a moderate deficit in EXT compared with the control group ($d = 0.63$ [0.06, 1.19]). The effect sizes revealed that the LAS coper group exhibited no differences in EXT compared with the CAI ($d = 0.43$ [-0.13, 0.98]) and control ($d = 0.08$ [-0.47, 0.64]) groups. The effect sizes revealed no differences in ABD between the CAI and LAS coper groups ($d = 0.25$ [-0.30, 0.80]), CAI and control groups

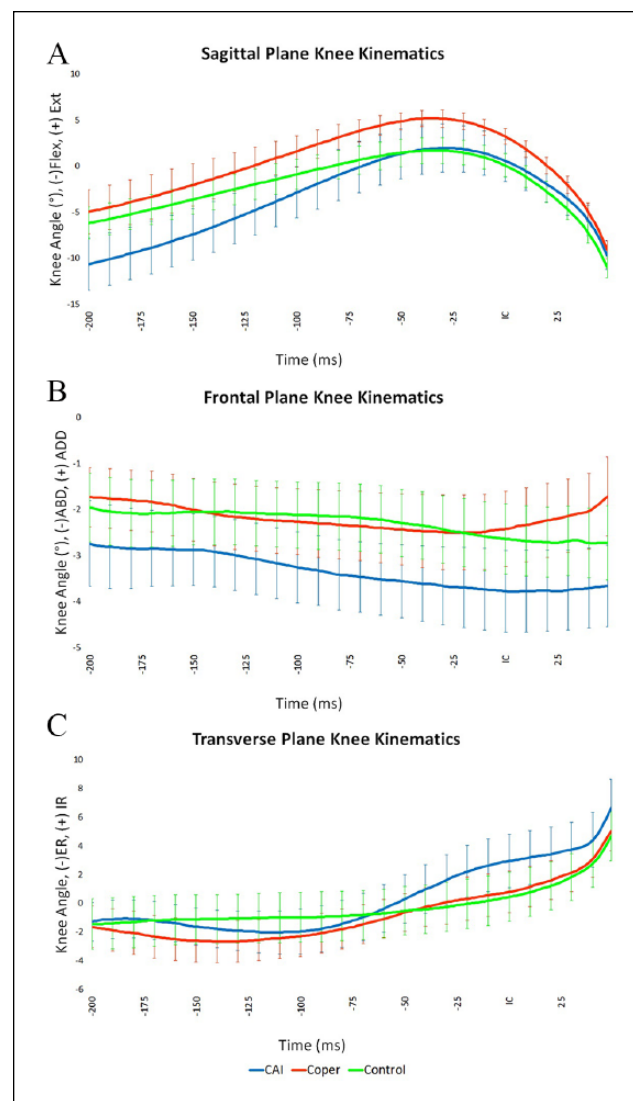


Figure 2. Pre- and postlanding knee kinematics of CAI, chronic ankle instability (CAI), lateral ankle sprain (LAS) coper, and control groups. Group means for knee kinematics in (A) sagittal, (B) frontal, and (C) transverse planes from 200 milliseconds pre-initial contact to 50 milliseconds post-initial contact. Error bars represent the standard error of the mean. No significant main effects existed for knee kinematics ($P > .05$). ABD, abduction; ADD, adduction; ER, external rotation; Ext, extension; Flex, flexion; IR, internal rotation.

($d = 0.57$ [0.00, 1.12]), or LAS coper and control groups ($d = 0.32$ [-0.24, 0.87]). We found no statistically significant correlations between hip strength and hip kinematics at each time interval (Table 3).

Discussion

The primary findings of this study were that participants with CAI demonstrated altered hip strength and landing

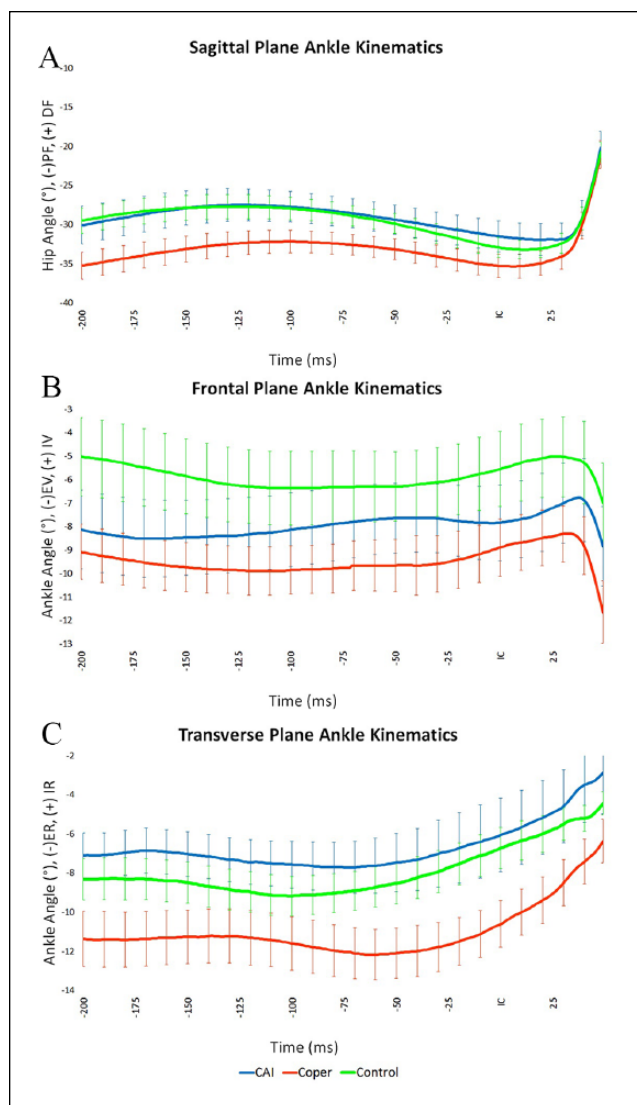


Figure 3. Pre- and postlanding ankle kinematics of CAI, chronic ankle instability (CAI), lateral ankle sprain (LAS) copers, and control groups. Group means for ankle kinematics in (A) sagittal, (B) frontal, and (C) transverse planes from 200 ms pre-initial contact to 50 ms post-initial contact. Error bars represent the standard error of the mean. No significant main effects existed for ankle kinematics ($P > .05$). DF, dorsiflexion; ER, external rotation; EV, eversion; IR, internal rotation; IV, inversion; PF, plantarflexion.

kinematics compared with LAS copers and controls. Specifically, participants with CAI exhibited less hip ABD before and after landing. The altered hip movement patterns of the CAI group potentially manifest from central nervous system changes following a peripheral joint injury and contribute to recurrent injury and CAI development.^{34,43} MacKinnon and Winter²⁹ explained that inversion moments at the subtalar joint may occur due, in part, to medial foot positioning and a subsequent increase in plantar pressure under the lateral aspect of the foot. Since our 3 groups did

not differ in 3D knee and ankle kinematics, the CAI group's decreased hip ABD angle might have translated to similar variations in medial-lateral foot placement, but future investigation is needed to verify this relationship. Others have examined 3D hip kinematics during landing in CAI and non-CAI populations,^{7,10,13,14} but our study is the first to identify differences in the frontal plane.

Although we found no differences in sagittal and transverse plane hip kinematics, such comparisons have previously exhibited inconsistent findings in similar populations.^{6,7,10,13,14,17,22} Similarly, we failed to discover group differences in ankle and knee joint kinematics. While these results are in agreement with others' findings regarding ankle^{13,14,17,22} and knee^{6,10,13,14} kinematics, altered ankle dorsiflexion^{6,8,10} and knee flexion angles^{8,17,22} have been reported in some landing kinematic studies of CAI and non-CAI populations. The reason for the vast discrepancies in kinematic differences is unclear but may be related to a number of factors. Across studies, heterogeneity existed within the CAI classifications, which commonly results in varying functional impairments.⁴⁰ Variations in task demands and footwear use between studies may have also influenced the conflicting results. We must also note that limitations in the reliability of frontal and transverse plane kinematic analyses may have affected these results.^{11,27} Additionally, a number of options exist for the analysis of temporal kinematic data,¹⁴ and thus alternative methods may have elicited different findings. Future research should continue to examine this area, as it may be highly relevant to CAI rehabilitation.

In addition to altered frontal plane hip positioning, the CAI group demonstrated deficits in EXT and ER strength measures. As our participants did not have any existing hip pathology that would limit hip strength, this finding provides further support for the presence of centrally mediated neuromuscular alterations.⁴³ Furthermore, deficits in isometric hip strength have been reported previously in participants with CAI.^{15,30,31,33} We aimed to determine relationships between hip muscular strength and hip kinematics to derive potential rehabilitation strategies for correcting functional deficits. We hypothesized that decreased posterolateral hip muscular strength would be associated with lower degrees of hip flexion, ABD, and ER during landing. Thus, we expected the reduced hip ABD angles of our CAI group to be associated with reduced ABD strength. Fatigue of the hip abductors has previously resulted in reduced medial-lateral postural stability,²⁸ which could promote LAS mechanisms by shifting plantar pressure under the lateral column of the foot.^{4,12,28,29} However, the hip motion of all 3 groups was not significantly related to isometric hip muscular strength. This finding may indicate that static measures of hip muscular strength are poor predictors of dynamic hip motion. Lawrence et al²⁶ previously reported that isometric hip

Table 2. Between-Group Comparisons of Isometric Hip Muscular Strength.^a

	CAI (n = 26)	LAS Coper (n = 25)	Control (n = 25)	One-Way ANOVA P Value	
EXT, Nm/kg	1.28 ± 0.40	1.49 ± 0.57	1.53 ± 0.39	$F_{(2, 73)} = 2.250$	.113
ABD, Nm/kg	1.56 ± 0.48	1.68 ± 0.47	1.83 ± 0.47	$F_{(2, 73)} = 2.135$	.126
ER, Nm/kg	0.57 ± 0.14 ^b	0.66 ± 0.15	0.70 ± 0.16	$F_{(2, 73)} = 5.181$	.008

Abbreviations: ABD, abduction; ANOVA, analysis of variance; CAI, chronic ankle instability; ER, external rotation; EXT, extension; LAS, lateral ankle sprain; SD, standard deviation.

Bold indicates statistical significance.

^aData presented as mean ± SD, unless otherwise noted

^bStatistically different from LAS coper and control groups ($P < .05$).

Table 3. Pearson Product Moment Correlations.^a

Outcome Variable	Predictor Variable	r Range	r, Initial Contact
CAI			
Sagittal hip angle	EXT	−0.24, −0.05	−0.23
Frontal hip angle	ABD	−0.31, −0.03	−0.28
Transverse hip angle	ER	−0.14, 0.00	−0.14
LAS Coper			
Sagittal hip angle	EXT	−0.18, −0.03	−0.10
Frontal hip angle	ABD	−0.31, 0.01	−0.30
Transverse hip angle	ER	−0.35, −0.15	−0.34
Control			
Sagittal hip angle	EXT	−0.42, −0.18	−0.34
Frontal hip angle	ABD	−0.29, −0.19	−0.28
Transverse hip angle	ER	−0.35, −0.22	−0.28

Abbreviations: ABD, abduction; CAI, chronic ankle instability; ER, external rotation; EXT, extension; LAS, lateral ankle sprain.

^aCorrelation analyses were conducted between hip strength and kinematic position at each millisecond across the time series. We examined associations between sagittal plane kinematics and EXT, frontal plane kinematics and ABD, and transverse plane kinematics and ER. Due to the vast number of analyses, we have presented the range of correlation coefficients (r) and that at initial contact during landing. No correlations were statistically significant ($P > .05$).

strength did not affect frontal plane hip motion during a landing. Boling and Padua⁴ instead reported that increased eccentric hip peak torque was associated with a less adducted hip position during a landing task. Thus, dynamic measures of hip muscular function such as isotonic and isokinetic strength and muscular activation may represent hip movement alterations more accurately.

The lack of differences between LAS coper and control groups provides further evidence that a portion of individuals with a previous ankle sprain are able to avoid developing characteristics of CAI, as well as similar patterns of motor control. While LAS copers and controls do not perform identically in all clinical outcomes,³² similarities in 3D landing kinematics and hip strength indicate that these outcomes are likely pertinent to an individual's ability to avoid recurrent ankle sprains and feelings of instability. Whether LAS copers are exposed to more comprehensive neuromuscular rehabilitation after injury or if

they have intrinsic neuromuscular adaptations that prevent the onset of CAI remains unclear.

Clinical Application

The results of this study indicate that hip kinematic landing patterns may need to be corrected in patients with CAI. Particularly, promoting hip ABD motion might lower their susceptibility to potentially traumatic supination moments. However, our data indicate that measures of isometric hip strength are poor indicators of these altered landing kinematics. Landing from a jump involves dynamic motor control, so dynamic measures of muscular function such as activation or isotonic strength may be more representative of movement patterns. Although isometric hip strength may not be directly related to movement patterns in those with CAI, their hip muscular strength deficits compared with LAS coper and control groups indicate that they still need to be addressed during rehabilitation. Individuals with CAI often demonstrate other neuromuscular impairments such as reduced dynamic balance, which is related to reduced isometric hip strength in that population.³¹ Hip strengthening exercises have exhibited benefits for individuals with CAI,³⁹ although the best method for targeting these movement patterns during rehabilitation is currently unknown. Previous work has shown that augmented feedback strategies can supplement strength training protocols as a means of influencing lower extremity kinematics during functional tasks.²³ Future studies should continue to explore the impact of reduced hip muscular strength on functional performance deficits in those with CAI as well as determine the best methods for rehabilitation.

Limitations

Some limitations present in this study should be noted. First, the retrospective study design was unable to clarify when the CAI group's neuromuscular alterations arose. It remains possible that impairments existed prior to injury and were not necessarily a consequence of a LAS. We intended to enhance clinical applicability by using isometric strength testing, but it may have provided a poor estimate of the hip musculature's influence to hip kinematics.

Our methods of evaluating 3D ankle, knee, and hip joint motion are potentially limited by marker misplacement and soft tissue artifact, which can produce error in lower extremity kinematic data, particularly in frontal and transverse planes.^{11,27}

Conclusion

In conclusion, participants with CAI exhibited altered hip landing kinematics compared with LAS copers and control groups. The differing frontal plane landing patterns may increase the likelihood of recurrent LAS or episodes of giving way in those with CAI. The CAI group also displayed reductions in isometric hip muscular strength, but these measures were not associated with the CAI group's lower extremity kinematics. Future studies should explore the influence of other measures of hip muscular function on landing kinematics and determine the best methods for correcting detected deficits.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article. ICMJE forms for all authors are available online.

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References

1. Anandacoomarasamy A, Barnsley L. Long term outcomes of inversion ankle injuries. *Br J Sports Med*. 2005;39(3):e14.
2. Bell AL, Pedersen DR, Brand RA. A comparison of the accuracy of several hip center location prediction methods. *J Biomech*. 1990;23(6):617-621.
3. Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J R Stat Soc Series B Stat Methodol*. 1995;57(1):289-300.
4. Boling M, Padua D. Relationship between hip strength and trunk, hip, and knee kinematics during a jump-landing task in individuals with patellofemoral pain. *Int J Sports Phys Ther*. 2013;8(5):661-669.
5. Braun BL. Effects of ankle sprain in a general clinic population 6 to 18 months after medical evaluation. *Arch Fam Med*. 1999;8(2):143-148.
6. Brown C, Padua D, Marshall SW, Guskiewicz K. Individuals with mechanical ankle instability exhibit different motion patterns than those with functional ankle instability and ankle sprain copers. *Clin Biomech (Bristol, Avon)*. 2008;23(6):822-831.
7. Brown CN, Padua DA, Marshall SW, Guskiewicz KM. Hip kinematics during a stop-jump task in patients with chronic ankle instability. *J Athl Train*. 2011;46(5):461-467.
8. Caulfield BM, Garrett M. Functional instability of the ankle: differences in patterns of ankle and knee movement prior to and post landing in a single leg jump. *Int J Sports Med*. 2002;23(1):64-68.
9. Davis RB, Ounpuu S, Tyburski D, Gage JR. A gait analysis data collection and reduction technique. *Hum Mov Sci*. 1991;10:575-587.
10. Delahunt E, Monaghan K, Caulfield B. Changes in lower limb kinematics, kinetics, and muscle activity in subjects with functional instability of the ankle joint during a single leg drop jump. *J Orthop Res*. 2006;24(10):1991-2000.
11. Della Croce U, Leardini A, Chiari L, Cappozzo A. Human movement analysis using stereophotogrammetry. Part 4: assessment of anatomical landmark misplacement and its effects on joint kinematics. *Gait Posture*. 2005;21(2):226-237.
12. de Ridder R, Witvrouw E, Dolphens M, Roosen P, Van Ginckel A. Hip strength as an intrinsic risk factor for lateral ankle sprains in youth soccer players: a 3-season prospective study. *Am J Sports Med*. 2017;45(2):410-416.
13. Doherty C, Bleakley C, Hertel J, Caulfield B, Ryan J, Delahunt E. Single-leg drop landing movement strategies in participants with chronic ankle instability compared with lateral ankle sprain 'copers'. *Knee Surg Sports Traumatol Arthrosc*. 2016;24(4):1049-1059.
14. Doherty C, Bleakley C, Hertel J, et al. Coordination and symmetry patterns during the drop vertical jump in people with chronic ankle instability and lateral ankle sprain copers. *Phys Ther*. 2016;96(8):1152-1161.
15. Friel K, McLean N, Myers C, Caceres M. Ipsilateral hip abductor weakness after inversion ankle sprain. *J Athl Train*. 2006;41(1):74-78.
16. Fuller EA. Center of pressure and its theoretical relationship to foot pathology. *J Am Podiatr Med Assoc*. 1999;89(6):278-291.
17. Gribble P, Robinson R. Differences in spatiotemporal landing variables during a dynamic stability task in subjects with CAI. *Scand J Med Sci Sports*. 2010;20(1):e63-e71.
18. Gribble PA, Bleakley CM, Caulfield BM, et al. 2016 consensus statement of the International Ankle Consortium: prevalence, impact and long-term consequences of lateral ankle sprains. *Br J Sports Med*. 2016;50(24):1493-1495.
19. Gribble PA, Delahunt E, Bleakley C, et al. Selection criteria for patients with chronic ankle instability in controlled research: a position statement of the International Ankle Consortium. *Br J Sports Med*. 2014;48(13):1014-1018.
20. Gribble PA, Delahunt E, Bleakley C, et al. Selection criteria for patients with chronic ankle instability in controlled research: a position statement of the International Ankle Consortium. *J Orthop Sports Phys Ther*. 2013;43(8):585-591.
21. Gribble PA, Delahunt E, Bleakley CM, et al. Selection criteria for patients with chronic ankle instability in controlled research: a position statement of the International Ankle Consortium. *J Athl Train*. 2014;49(1):121-127.
22. Gribble PA, Robinson RH. Alterations in knee kinematics and dynamic stability associated with chronic ankle instability. *J Athl Train*. 2009;44(4):350-355.
23. Herman DC, Onate JA, Weinhold PS, et al. The effects of feedback with and without strength training on lower extremity biomechanics. *Am J Sports Med*. 2009;37(7):1301-1308.
24. Hopkins JT, Coglianese M, Glasgow P, Reese S, Seeley MK. Alterations in evtor/invertor muscle activation and center of

- pressure trajectory in participants with functional ankle instability. *J Electromyogr Kinesiol*. 2012;22(2):280-285.
25. Koldenhoven RM, Feger MA, Fraser JJ, Saliba S, Hertel J. Surface electromyography and plantar pressure during walking in young adults with chronic ankle instability. *Knee Surg Sports Traumatol Arthrosc*. 2016;24(4):1060-1070.
26. Lawrence RK III, Kernozek TW, Miller EJ, Torry MR, Reuteman P. Influences of hip external rotation strength on knee mechanics during single-leg drop landings in females. *Clin Biomech (Bristol, Avon)*. 2008;23(6):806-813.
27. Leardini A, Chiari L, Della Croce U, Cappozzo A. Human movement analysis using stereophotogrammetry. Part 3. Soft tissue artifact assessment and compensation. *Gait Posture*. 2005;21(2):212-225.
28. Lee SP, Powers C. Fatigue of the hip abductors results in increased medial-lateral center of pressure excursion and altered peroneus longus activation during a unipedal landing task. *Clin Biomech (Bristol, Avon)*. 2013;28(5):524-529.
29. MacKinnon CD, Winter DA. Control of whole body balance in the frontal plane during human walking. *J Biomech*. 1993;26(6):633-644.
30. McCann RS, Bolding BA, Terada M, Kosik KB, Crossett ID, Gribble PA. Isometric hip strength and dynamic stability of individuals with chronic ankle instability. *J Athl Train*. 2018;53(7):672-678.
31. McCann RS, Crossett ID, Terada M, Kosik KB, Bolding BA, Gribble PA. Hip strength and star excursion balance test deficits of patients with chronic ankle instability. *J Sci Med Sport*. 2017;20:992-996.
32. McCann RS, Terada M, Kosik KB, Gribble PA. Energy dissipation differs between females with and without chronic ankle instability. *Scand J Med Sci Sports*. 2018;28(3):1227-1234.
33. Nicholas JA, Strizak AM, Veras G. A study of thigh muscle weakness in different pathological states of the lower extremity. *Am J Sports Med*. 1976;4(6):241-248.
34. Pietrosimone BG, McLeod MM, Lepley AS. A theoretical framework for understanding neuromuscular response to lower extremity joint injury. *Sport Health*. 2012;4(1):31-35.
35. Plante JE, Wikstrom EA. Differences in clinician-oriented outcomes among controls, copers, and chronic ankle instability groups. *Phys Ther Sport*. 2013;14(4):221-226.
36. Ross SE, Guskiewicz KM. Examination of static and dynamic postural stability in individuals with functionally stable and unstable ankles. *Clin J Sport Med*. 2004;14(6):332-338.
37. Shah S, Thomas AC, Noone JM, Blanchette CM, Wikstrom EW. Incidence and cost of ankle sprains in United States emergency departments. *Sport Health*. 2016;8(6):547-552.
38. Sinclair J, Hebron J, Taylor PJ. The test-retest reliability of knee joint center location techniques. *J Appl Biomech*. 2015;31(2):117-121.
39. Smith BI, Curtis D, Docherty CL. Effects of hip strengthening on neuromuscular control, hip strength, and self-reported functional deficits in individuals with chronic ankle instability. *J Sport Rehabil*. 2018;27(4):364-370.
40. Terada M, Bowker S, Hiller CE, Thomas AC, Pietrosimone B, Gribble PA. Quantifying levels of function between different subgroups of chronic ankle instability. *Scand J Med Sci Sports*. 2017;27(2):650-660.
41. Thewlis D, Richards J, Bower J. Discrepancies in knee joint moments using common anatomical frames defined by different palpable landmarks. *J Appl Biomech*. 2008;24(2):185-190.
42. Thorborg K, Petersen J, Magnusson SP, Holmich P. Clinical assessment of hip strength using a hand-held dynamometer is reliable. *Scand J Med Sci Sports*. 2010;20(3):493-501.
43. Ward S, Pearce AJ, Pietrosimone B, Bennell K, Clark R, Bryant AL. Neuromuscular deficits following peripheral joint injury: a neurophysiological hypothesis. *Muscle Nerve*. 2015;51(3):327-332.
44. Waterman BR, Owens BD, Davey S, Zacchilli MA, Belmont PJ Jr. The epidemiology of ankle sprains in the United States. *J Bone Joint Surg Am*. 2010;92(13):2279-2284.
45. Whiting WC, Zernicke RF. *Biomechanics of Musculoskeletal Injury*. 2nd ed. Champaign, IL: Human Kinetics; 2008.
46. Wikstrom EA, Brown CN. Minimum reporting standards for copers in chronic ankle instability research. *Sports Med*. 2014;44(2):251-268.